

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0603

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I D.Sindhu (192424197) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

D. Sindhu.

Signature of the student:

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

(a) Describe the steps involved in training, validating, and evaluating the model.

(b) Explain how the system can assist in early warning and emergency response.

(5 Marks)

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example.

(5 Marks)

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis.
(5 Marks)

Dataset:

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

6. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only.
(5 Marks)

Dataset:

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

7. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples.
(5 Marks)

Dataset:

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
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1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

8. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the **specific (S)** and **general (G)** boundaries. Show step-by-step updates after each training example. (5 Marks)

Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

9. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the **version space (S and G boundaries)** and show the updates after each example. (5 Marks)

Dataset:

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No

10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions. (5 Marks)

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples. (5 Marks)

12. Design a machine learning model for spam email classification using labeled data. (5 Marks)

- (a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).
- (b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.
- (c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: (5 Marks)

Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept *EnjoySport*.

- (a) Explain the initialization of S (specific boundary) and G (general boundary).
- (b) Show how S and G change with each training example.
- (c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)

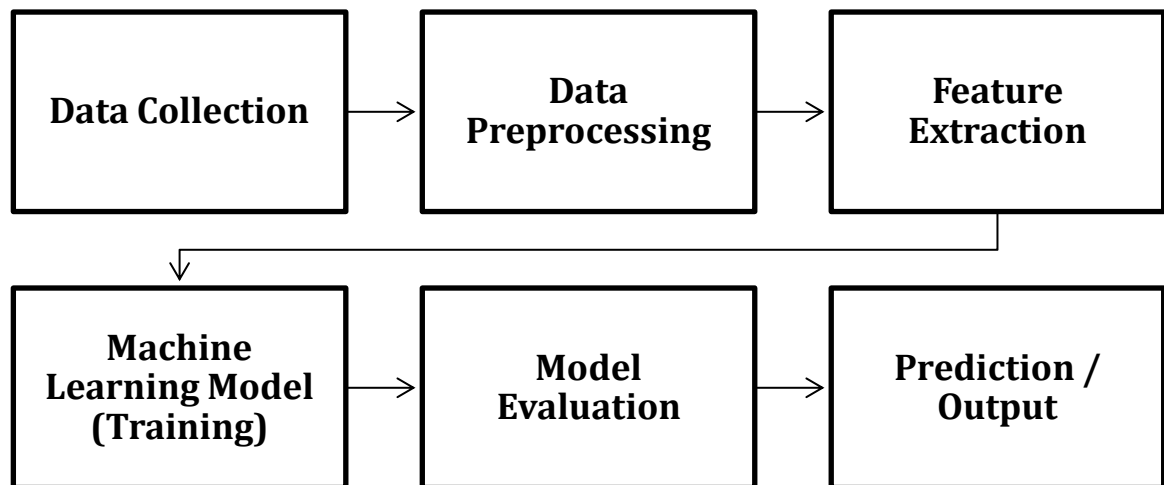
- (a) Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest).
- (b) Explain feature scaling and handling missing data.
- (c) Describe how overfitting can be detected and avoided.

15. Explain the concept of clustering in unsupervised learning. (5 Marks)

- (a) Describe the working of the K-Means algorithm with a step-by-step example.
- (b) Explain how the value of K is chosen using the Elbow Method.
- (c) Compare K-Means with Hierarchical Clustering.

Answers:

1.Blocks in a Machine Learning System



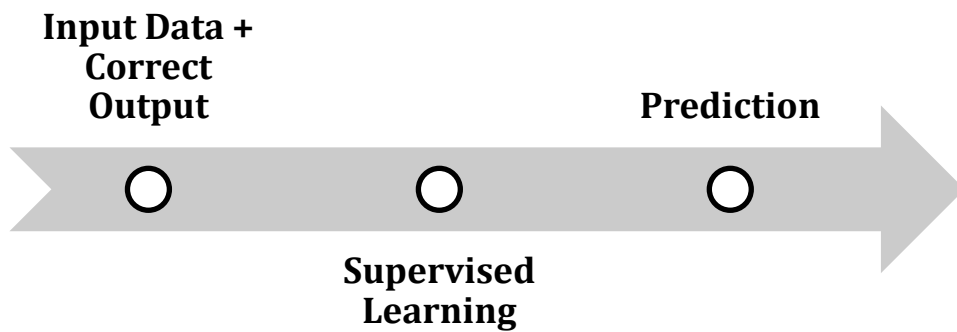
Explanation of the Blocks

- 1. Data Collection**
 - Collect data from different sources.
 - **Example:** Student marks data.
- 2. Data Preprocessing**
 - Clean the data by removing errors and missing values.
 - Makes the data ready for learning.
- 3. Feature Extraction**
 - Select important information (features) from the data.
 - **Example:** Attendance and marks.
- 4. Machine Learning Model**
 - Train the model using a machine learning algorithm.
 - The model learns patterns from the data.
- 5. Model Evaluation**
 - Check how well the model works using test data.
 - **Example:** Measure accuracy.
- 6. Prediction / Output**
 - The trained model predicts the result for new data.
 - **Example:** Predict whether a student will pass or fail.

Supervised Learning

Definition:

Supervised learning uses **labelled data**, where the correct output is already known.



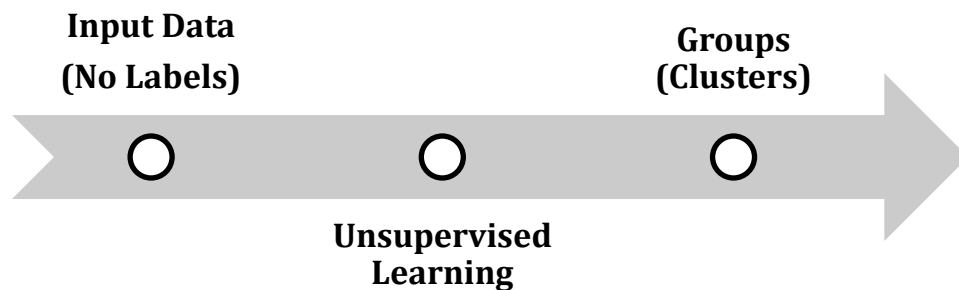
Example:

Predicting whether a student will Pass or Fail based on study hours and marks.

Unsupervised Learning

Definition:

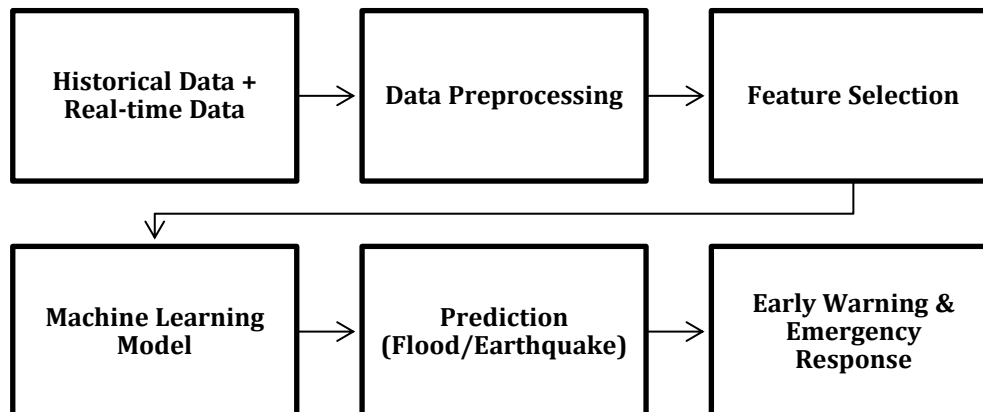
Unsupervised learning uses **unlabelled data**. It finds similar groups or patterns in the data.



Example:

Grouping customers into different categories based on their shopping habits.

2. Machine Learning System for Disaster Management



Choice of Model

- Random Forest or Decision Tree can be used because they provide accurate predictions and handle large datasets.
- Example: Predicting flood risk using rainfall and river water level.

Data Preprocessing

- Remove missing and duplicate data.
- Normalize the data.
- Convert data into a suitable format.

Feature Selection

Select important features such as:

- Rainfall
- River water level
- Temperature
- Earthquake magnitude
- Seismic activity

(a) Steps in Training, Validation, and Evaluation

1. **Training:** Train the model using historical disaster data.
2. **Validation:** Test the model with validation data and adjust if needed.
3. **Evaluation:** Measure performance using Accuracy, Precision, and Recall.

(b) Early Warning and Emergency Response

- Predicts floods and earthquakes before they occur.
- Sends early warning alerts to people and authorities.
- Helps emergency teams plan rescue operations.
- Reduces loss of life and property.

3. Find-S Algorithm

Step 1: Initialize the Hypothesis

The initial hypothesis is the most specific.

$H_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Consider only positive examples (Yes) and ignore negative examples.

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

Step 2: Update the Hypothesis

Positive Example 3

(Overcast, Hot, High, Weak)

$H_1 = \langle \text{Overcast, Hot, High, Weak} \rangle$

Positive Example 4

(Rainy, Mild, High, Weak)

Compare with H_1 :

- Outlook: Overcast $\neq$ Rainy $\rightarrow$?
- Temperature: Hot $\neq$ Mild $\rightarrow$?
- Humidity: High = High $\rightarrow$ High
- Wind: Weak = Weak $\rightarrow$ Weak

$H_2 = \langle ?, ?, \text{High, Weak} \rangle$

Positive Example 5

(Rainy, Cool, Normal, Weak)

Compare with H_2 :

- Outlook: ? $\rightarrow$?
- Temperature: ? $\rightarrow$?
- Humidity: High $\neq$ Normal $\rightarrow$?
- Wind: Weak = Weak $\rightarrow$ Weak

$H_3 = \langle ?, ?, ?, \text{Weak} \rangle$

Positive Example 7

(Overcast, Mild, Normal, Strong)

Compare with H_3 :

- Outlook: ? $\rightarrow$?
- Temperature: ? $\rightarrow$?
- Humidity: ? $\rightarrow$?
- Wind: Weak $\neq$ Strong $\rightarrow$?

$H_4 = \langle ?, ?, ?, ? \rangle$

Final Maximally Specific Hypothesis

$H = \langle ?, ?, ?, ? \rangle$

Answer

- Initial Hypothesis: $\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
- After Example 3: $\langle \text{Overcast}, \text{Hot}, \text{High}, \text{Weak} \rangle$
- After Example 4: $\langle ?, ?, \text{High}, \text{Weak} \rangle$
- After Example 5: $\langle ?, ?, ?, \text{Weak} \rangle$
- After Example 7: $\langle ?, ?, ?, ? \rangle$

Final Hypothesis:

$\langle ?, ?, ?, ? \rangle$

Interpretation: Any value of Outlook, Temperature, Humidity, and Wind is acceptable because no single attribute is common across all positive training examples.

4. Candidate Elimination Algorithm (5 Marks)

Step 1: Initialize Boundaries

- Specific Boundary (S) = $\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
- General Boundary (G) = $\langle ?, ?, ?, ? \rangle$

Step 2: Process Each Training Example

Example 1

(Sunny, Warm, Normal, Weak) $\rightarrow$ Yes

Update S:

$S = \langle \text{Sunny}, \text{Warm}, \text{Normal}, \text{Weak} \rangle$

$G = \langle ?, ?, ?, ? \rangle$

Example 2

(Sunny, Warm, High, Strong) $\rightarrow$ Yes

Generalize S:

- Weather: Sunny = Sunny $\rightarrow$ Sunny
- Temperature: Warm = Warm $\rightarrow$ Warm
- Humidity: Normal $\neq$ High $\rightarrow ?$
- Wind: Weak $\neq$ Strong $\rightarrow ?$

$S = \langle \text{Sunny}, \text{Warm}, ?, ? \rangle$

$G = \langle ?, ?, ?, ? \rangle$

Example 3

(Rainy, Cold, High, Weak) $\rightarrow$ No

Specialize G to exclude this negative example while remaining consistent with S.

$G = \langle \text{Sunny}, ?, ?, ? \rangle$

$S = \langle \text{Sunny}, \text{Warm}, ?, ? \rangle$

Example 4

(Sunny, Cold, Normal, Weak) → Yes

Generalize S:

- Weather: Sunny = Sunny → Sunny
- Temperature: Warm ≠ Cold → ?
- Humidity: ? → ?
- Wind: ? → ?

S = ⟨Sunny, ?, ?, ?⟩

G = ⟨Sunny, ?, ?, ?⟩

Example 5

(Rainy, Warm, Normal, Strong) → No

This negative example is already excluded because Weather = Rainy.

So, S and G remain unchanged.

S = ⟨Sunny, ?, ?, ?⟩

G = ⟨Sunny, ?, ?, ?⟩

Final Answer

- Specific Boundary (S): ⟨Sunny, ?, ?, ?⟩
- General Boundary (G): ⟨Sunny, ?, ?, ?⟩

Version Space = {⟨Sunny, ?, ?, ?⟩}

This means the person is predicted to enjoy the sport whenever the weather is Sunny, regardless of temperature, humidity, or wind.

5. Find-S Algorithm (5 Marks)

Step 1: Initialize the Hypothesis

The initial hypothesis is the most specific.

$H_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Consider only positive examples (Yes) and ignore all negative examples.

Dataset

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

Step 2: Update the Hypothesis

Positive Example 3

(Middle, High, No, Fair)

$H_1 = \langle \text{Middle, High, No, Fair} \rangle$

Positive Example 4

(Senior, Medium, No, Fair)

Compare with H_1 :

- Age: Middle $\neq$ Senior $\rightarrow ?$
- Income: High $\neq$ Medium $\rightarrow ?$
- Student: No = No $\rightarrow$ No
- Credit Rating: Fair = Fair $\rightarrow$ Fair

$H_2 = \langle ?, ?, \text{No}, \text{Fair} \rangle$

Final Maximally Specific Hypothesis

$H = \langle ?, ?, \text{No}, \text{Fair} \rangle$

Answer

Step	Hypothesis
Initial	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
After Example 3	$\langle \text{Middle}, \text{High}, \text{No}, \text{Fair} \rangle$
After Example 4	$\langle ?, ?, \text{No}, \text{Fair} \rangle$

Final Hypothesis:

$\langle ?, ?, \text{No}, \text{Fair} \rangle$

Interpretation: A customer is likely to buy the product if:

- Student = No
- Credit Rating = Fair
- Age can be any value (?)
- Income can be any value (?)

6. Find-S Algorithm

Step 1: Initialize the Hypothesis

The initial hypothesis is the most specific.

$H_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Consider only positive examples (Yes) and ignore all negative examples.

Dataset

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

Step 2: Update the Hypothesis

Positive Example 2

(Rainy, Mild, Evening, Happy)

$H_1 = \langle \text{Rainy}, \text{Mild}, \text{Evening}, \text{Happy} \rangle$

Positive Example 3

(Sunny, Mild, Evening, Happy)

Compare with H_1 :

- Weather: Rainy $\neq$ Sunny $\rightarrow$?
- Temperature: Mild = Mild $\rightarrow$ Mild
- Time: Evening = Evening $\rightarrow$ Evening
- Mood: Happy = Happy $\rightarrow$ Happy

$H_2 = \langle ?, \text{Mild}, \text{Evening}, \text{Happy} \rangle$

Positive Example 4

(Cloudy, Cool, Morning, Happy)

Compare with H_2 :

- Weather: ? $\rightarrow$?
- Temperature: Mild $\neq$ Cool $\rightarrow$?
- Time: Evening $\neq$ Morning $\rightarrow$?
- Mood: Happy = Happy $\rightarrow$ Happy

$H_3 = \langle ?, ?, ?, \text{Happy} \rangle$

Final Maximally Specific Hypothesis

$H = \langle ?, ?, ?, \text{Happy} \rangle$

Answer

Step	Hypothesis
Initial	$\langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$
After Example 2	$\langle \text{Rainy}, \text{Mild}, \text{Evening}, \text{Happy} \rangle$
After Example 3	$\langle ?, \text{Mild}, \text{Evening}, \text{Happy} \rangle$
After Example 4	$\langle ?, ?, ?, \text{Happy} \rangle$

Final Hypothesis:

$\langle ?, ?, ?, \text{Happy} \rangle$

Interpretation: A person is likely to go for a workout when the Mood is Happy, regardless of Weather, Temperature, or Time.

7. Find-S Algorithm

Step 1: Initialize

$H_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Ignore negative examples.

Positive Example	Hypothesis
Example 1: (Employed, High, Good, Clean)	$\langle \text{Employed}, \text{High}, \text{Good}, \text{Clean} \rangle$
Example 2: (Employed, Medium, Good, Clean)	$\langle \text{Employed}, ?, \text{Good}, \text{Clean} \rangle$
Example 4: (Employed, Medium, Excellent, Clean)	$\langle \text{Employed}, ?, ?, \text{Clean} \rangle$

Final Hypothesis

$\langle \text{Employed}, ?, ?, \text{Clean} \rangle$

8. Candidate Elimination Algorithm (5 Marks)

Initial

$S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

$G = \langle ?, ?, ?, ? \rangle$

Step	S Boundary	G Boundary
Example 1 (Yes)	$\langle \text{Young, High, Frequent, Mobile} \rangle$	$\langle ?, ?, ?, ? \rangle$
Example 2 (Yes)	$\langle \text{Young, High, ?, Mobile} \rangle$	$\langle ?, ?, ?, ? \rangle$
Example 3 (No)	$\langle \text{Young, High, ?, Mobile} \rangle$	$\langle \text{Young, ?, ?, ?} \rangle$
Example 4 (Yes)	$\langle \text{Young, ?, ?, Mobile} \rangle$	$\langle \text{Young, ?, ?, ?} \rangle$
Example 5 (No)	No change	$\langle \text{Young, ?, ?, ?} \rangle$

Final Answer

- $S = \langle \text{Young, ?, ?, Mobile} \rangle$
- $G = \langle \text{Young, ?, ?, ?} \rangle$

9. Candidate Elimination Algorithm (5 Marks)

Initial

$S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

$G = \langle ?, ?, ?, ? \rangle$

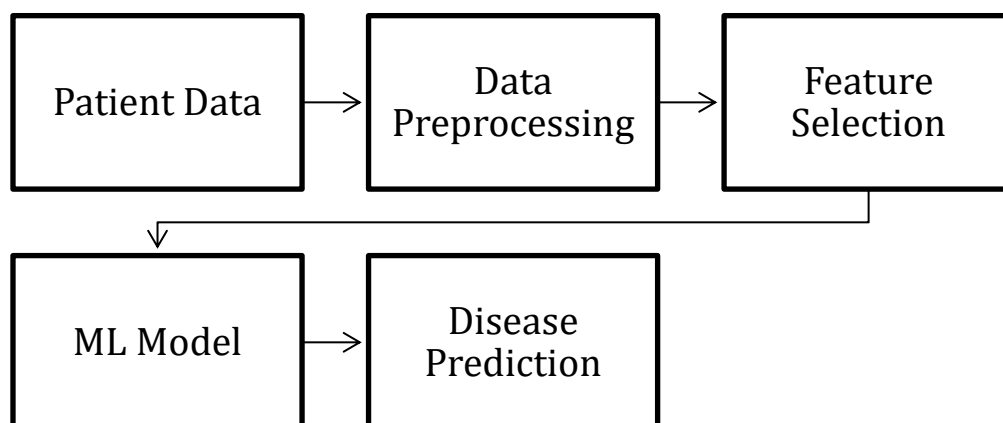
Step	S Boundary	G Boundary
Example 1 (Yes)	$\langle \text{Balanced, Regular, Adequate, Low} \rangle$	$\langle ?, ?, ?, ? \rangle$
Example 2 (Yes)	$\langle \text{Balanced, Regular, ?, ?} \rangle$	$\langle ?, ?, ?, ? \rangle$
Example 3 (No)	$\langle \text{Balanced, Regular, ?, ?} \rangle$	$\langle \text{Balanced, ?, ?, ?} \rangle$
Example 4 (Yes)	$\langle \text{Balanced, ?, ?, ?} \rangle$	$\langle \text{Balanced, ?, ?, ?} \rangle$
Example 5 (No)	No change	$\langle \text{Balanced, ?, ?, ?} \rangle$

Final Answer

- $S = \langle \text{Balanced, ?, ?, ?} \rangle$
- $G = \langle \text{Balanced, ?, ?, ?} \rangle$

10. Machine Learning System for Health Analysis

Sketch



Algorithm

- Decision Tree
- Random Forest
- Logistic Regression

(Random Forest is suitable because it gives good accuracy.)

Data Preprocessing

- Remove missing values
- Normalize data
- Remove duplicates

Feature Selection

Use important features:

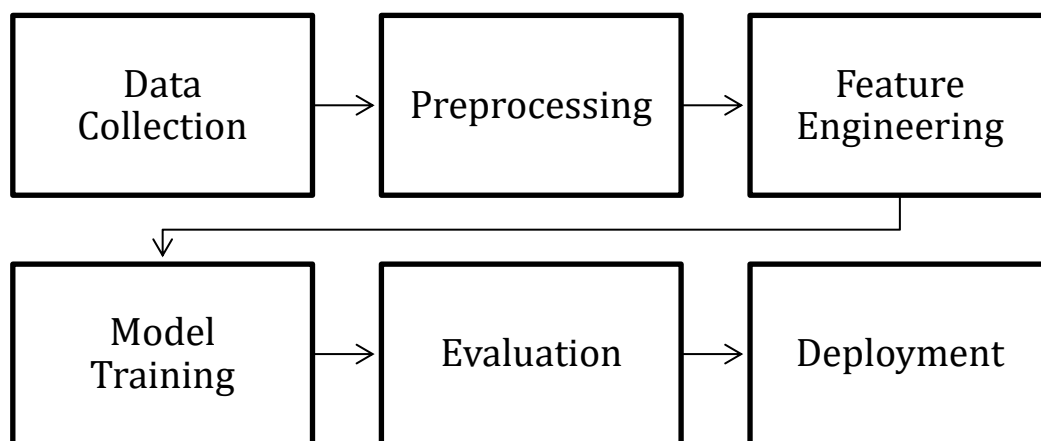
- Age
- Blood Pressure
- Glucose Level
- Lifestyle

Model Training

- Split data into Training and Testing.
- Train the model.
- Evaluate accuracy.
- Predict heart disease or diabetes.

11. Machine Learning Pipeline

Diagram



Stages

- **Data Collection:** Collect data.
- **Preprocessing:** Clean data.
- **Feature Engineering:** Select useful features.
- **Training:** Train model.
- **Evaluation:** Measure accuracy.
- **Deployment:** Use model for prediction.

Training Error

Error on training data.

Example: 95% accuracy on training data.

Generalization Error

Error on new unseen data.

Example: 85% accuracy on testing data.

12. Spam Email Classification

(a) Algorithm

Naïve Bayes

Reason:

- Fast
- Simple
- Best for text classification

(b) Preprocessing

- Tokenization
- Remove stop words
- Convert text into vectors (TF-IDF)

(c) Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score

13. Candidate Elimination Algorithm

(a) Initialization

- $S = \langle \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset \rangle$
- $G = \langle ?, ?, ?, ?, ?, ? \rangle$

(b) Updates

- Positive examples $\rightarrow$ Generalize **S**
- Negative examples $\rightarrow$ Specialize **G**

(c) Find-S vs Candidate Elimination

Find-S	Candidate Elimination
Uses only positive examples	Uses positive and negative examples
Finds one hypothesis	Finds complete version space
Simple	More accurate

14. House Price Prediction (5 Marks)

(a) Regression Model

Random Forest Regression

Reason:

- High accuracy
- Handles complex data

(b) Feature Scaling & Missing Data

- Normalize features

- Replace missing values using mean/median

(c) Avoid Overfitting

- Cross Validation
- Pruning
- Train/Test Split

15. Clustering in Unsupervised Learning (5 Marks)

Definition

Clustering groups similar data without labels.

Example:

Customer segmentation.

(a) K-Means Algorithm

Steps:

1. Choose K clusters.
2. Select initial centroids.
3. Assign data to nearest centroid.
4. Update centroids.
5. Repeat until clusters do not change.

(b) Elbow Method

- Try different values of **K**.
- Plot **K** vs **WCSS (Within-Cluster Sum of Squares)**.
- Choose the point where the graph forms an **elbow**.

(c) K-Means vs Hierarchical Clustering

K-Means	Hierarchical
Need K value	No need to specify K initially
Faster	Slower
Uses centroids	Uses dendrogram
Suitable for large datasets	Suitable for small datasets

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand